Açı modülasyonu

$$y(t) = A_c \cos\underbrace{(2\pi f_c t + \phi(t))}_{\theta(t)}$$

↳ $k_p m(t)$ FZM PM

$= 2\pi k_f \int_{-\infty}^{t} m(z)dz$ FRM

$$f_i(t) = \frac{1}{2\pi}\frac{d\theta(t)}{dt} = f_c + \frac{1}{2\pi}\frac{d\phi(t)}{dt}$$

$$\theta(t) = 2\pi \int_{-\infty}^{t} f_i(z)dz$$

$$P_y = \frac{A_c^2}{2}$$

$BG = \infty$ (Teorik)

Carson kuralı

(Açı modülasyonun etkin BG)

$BG\{m(t)\} = B$ Hz

$BG = 2B(\beta+1)$ Hz , β ↓ sapma oranı, $\beta = \frac{\Delta f}{B}$

FM(FRM) $f_i(t) = f_c + k_f m(t) \rightarrow \Delta f = k_f \cdot m_p$, $m_p = maks|m(t)|$

PM(FZM) $f_i(t) = f_c + \frac{k_p}{2\pi}\dot{m}(t) \rightarrow \Delta f = \frac{k_p}{2\pi} m_p'$, $m_p' = maks|\dot{m}(t)|$ ($\dot{m}$: t'ye göre türev)

$\Delta\phi = maks|\phi(t)|$

$k_f, k_p << \rightarrow \Delta f << \rightarrow \beta << 1 \Rightarrow BG = 2B$ NBFM / NBPM

$k_f, k_p >> \rightarrow \Delta f >> \rightarrow \beta >> 1 \Rightarrow BG \cong 2\Delta f$ WBFM / WBPM

Örnek

$m(t) = a\cos(2\pi f_m t)$

$c(t) = A_c \cos(2\pi f_c t)$

PM: $\phi(t) = k_p m(t) = \frac{k_p \cdot a}{\beta}\cos(2\pi f_m t) = \beta\cos(2\pi f_m t)$ ($\beta = \frac{\Delta f}{B}$)

FM: $\phi(t) = 2\pi k_f \int m(z)dz = 2\pi k_f \int a\cos(2\pi f_m t)dt$

$= 2\pi k_f \frac{a\sin(2\pi f_m t)}{2\pi f_m} = \frac{k_f \cdot a}{f_m}\cdot\sin(2\pi f_m t)$ ($k_f \cdot m_p$ / β)

$= \beta \sin(2\pi f_m t)$

$$y(t) = \begin{cases} A_c \cos(2\pi f_c t + \beta \cos(2\pi f_m t)) & PM \\ A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t)) & FM \end{cases}$$

Saf sinüsoidal işaretin açı modülasyonu

$m(t) = a\cos(2\pi f_m t) \Rightarrow y_{FM}(t) = A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t)), \beta = \frac{k_f \cdot a}{f_m}$

$m(t) = a\sin(2\pi f_m t) \Rightarrow y_{PM}(t) = A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t)), \beta = k_p \cdot a$

$$\hat{y}(t) = A_c e^{j(2\pi f_c t + \beta \sin(2\pi f_m t))} = A_c e^{j2\pi f_c t} \cdot \underbrace{e^{j\beta \sin(2\pi f_m t)}}_{x(t)}$$

$y(t) = Re\{y(t)\}$

$x(t) = \underbrace{e^{j\beta \sin(2\pi f_m t)}}_{T_m = \frac{1}{f_m} \text{ ile periyodiktir}} = \sum_{n=-\infty}^{\infty} x_n e^{j2\pi n f_m t}$

$$x_n = \frac{1}{T_m} \int_{T_m} x(t) e^{-j2\pi n f_m t} dt = \frac{1}{T_m} \int_0^{T_m} e^{j\beta \sin(2\pi f_m t)} \cdot e^{-j2\pi n f_m t} dt$$

$2\pi f_m t = \alpha \Rightarrow dt = \frac{d\alpha}{2\pi f_m}$

$$x_n = \frac{1}{T_m} \int_0^{2\pi} e^{j\beta \sin(\alpha)} \cdot e^{-jn\alpha} \frac{d\alpha}{2\pi f_m}$$

$$\Rightarrow x_n = \frac{1}{2\pi} \int_0^{2\pi} e^{j(\beta \sin\alpha - n\alpha)} d\alpha$$

$$x_n = \frac{1}{2\pi} \int_0^{2\pi} e^{j(\beta \sin\alpha - n\alpha)} d\alpha = J_n(\beta)$$

analitik olarak hesaplayamayız, nümerik olarak hesaplanır

Bessel function of the 1st kind and nth order
(n mertebeden 1 tür Bessel fonk)

$$x(t) = \sum_{n=-\infty}^{+\infty} J_n(\beta) e^{j2\pi n f_m t}$$

$$\hat{y}(t) = A_c e^{j2\pi f_c t} \sum_{n=-\infty}^{\infty} J_n(\beta) e^{j2\pi n f_m t}$$

$$\hat{y}(t) = A_c \sum_{n=-\infty}^{\infty} \underbrace{J_n(\beta)}_{\text{reel}} e^{j[2\pi (f_c + n f_m) t]}$$

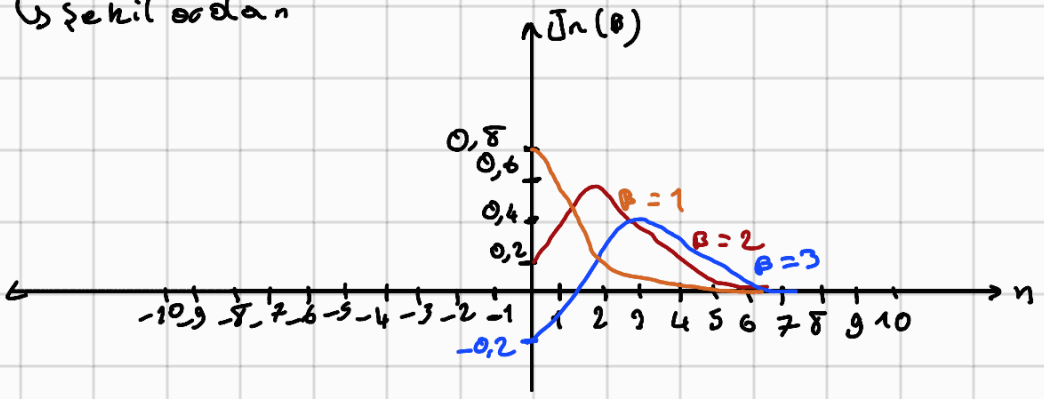
$$y(t) = Re\{\hat{y}(t)\} = A_c \sum_{n=-\infty}^{+\infty} J_n(\beta) \cdot \cos(2\pi(f_c + 2nf_m)t)$$

$$g(t) = A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t))$$
$$= A_c \sum_{n=-\infty}^{+\infty} J_n(\beta) \cos(2\pi(f_c + nf_m)t)$$

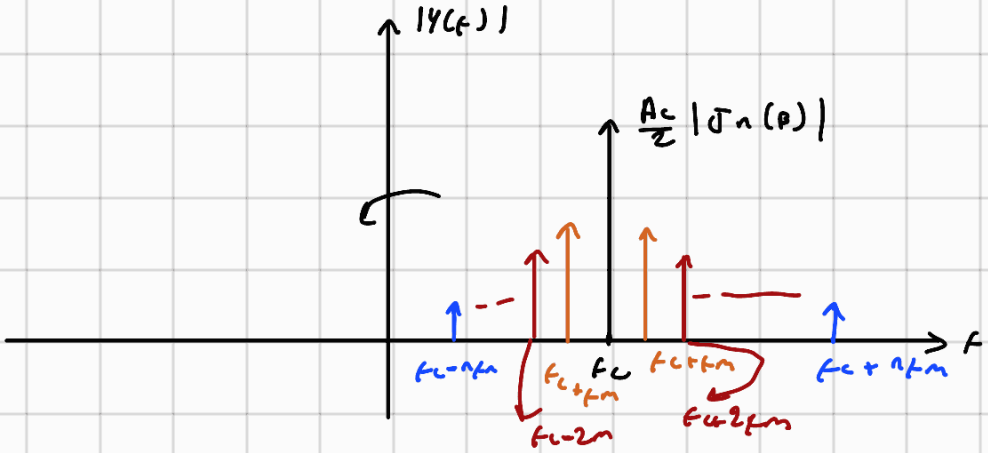
Bessel fonksiyonunu inceleyelim

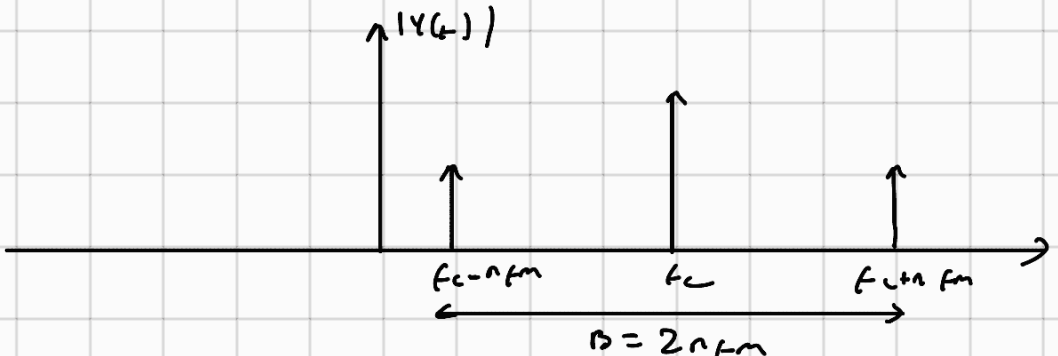
$$J_{-n}(\beta) = (-1)^n J_n(\beta)$$
$$|J_{-n}(\beta)| = |J_n(\beta)|$$

Lathi
(Prakis, s.176, Fig 4.4 benzeri)
şekilden



$n > \beta + 1$ için $J_n(\beta) << 0$





$= 2(\beta+1) f_m$, $B \in \{n(t)\} = f_n = B$

$= 2(\beta+1)B = 2B(\beta+1)$

Açı modülasyonu lineer değildir.

Örnek: (Proakis)

$c(t) = 10\cos(2\pi f_c t)$, $m(t) = \cos(20\pi t)$

FM with $k_f = 50$

Find the expression for the modulated signal and determine how many harmonics should be selected to contain 99 % of the modulated signal power.

$$\phi(t) = 2\pi k_f \int m(z)\,dz = \frac{2\pi k_f}{20\pi}\sin(20\pi t) = 5\sin(20\pi t)$$

β

$$y(t) = A_c \cos(2\pi f_c t + \phi(t))$$

$$y(t) = 10\cos(2\pi f_c t + \underbrace{5\sin(2\pi f_m t)}_{\phi(t)})$$

$$A_c = 10,\quad \beta = \frac{\Delta f}{B} = \frac{k_f \cdot m_p}{10} \overset{\downarrow 50,\ \downarrow 1}{=} 5$$

$$g(t) = 10\cos(2\pi f_c t + \underbrace{5\sin(20\pi t)}_{\text{periyodik}}) \rightarrow P_g = \frac{A_c^2}{2} = 50 \text{ watt } (\%100)$$

$$g(t) = A_c \sum_{n=-\infty}^{+\infty} J_n(\beta)\cos[2\pi(f_c + nf_m)t]$$

$$= 10 \sum_{n=-\infty}^{+\infty} J_n(5)\cos[2\pi(f_c + n10)t]$$

$$P_g = \frac{A_c^2}{2} \sum_{n=-\infty}^{+\infty} J_n^2(\beta)$$

$\frac{A_c^2}{2}$: güç (%100)

$\sum_{n=-\infty}^{+\infty} J_n^2(\beta) = 1$

$$\Rightarrow \sum_{n=-k}^{k} J_n^2(\beta) \geq 0{,}99$$

k tane harmonik

$$J_0^2(5) + 2\sum_{n=1}^{k} J_n^2(5) \geq 0{,}99$$

Sayfa 170, Tablo 4.1'e göre

Burada $k = 6$ için sağlanıyor

99 % of the power is within the frequencies $f_c \pm k \cdot 10 \rightarrow f_m$ } $f_c - 6\cdot 10$ —— $f_c + 6\cdot 10$

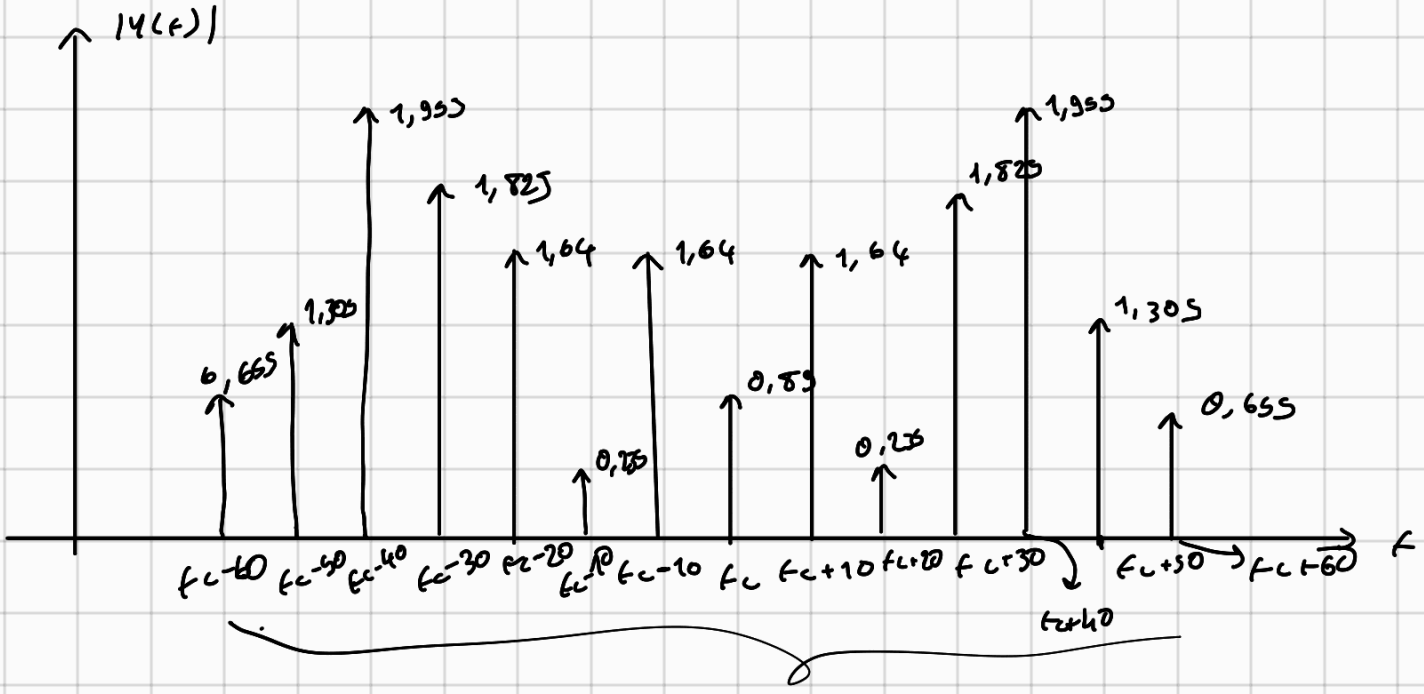
B Gerektiği = $2 \times 6 \times 10 = 120$ Hz

$2(\beta+1)\cdot f_m$

$2(\beta+1)B$ V

$\frac{A_c}{2} = 5$ $J_n(\beta)$

n	$J_n(\beta)$	
0	-0,178	→ -0,89
1	-0,328	→ -1,64
2	0,045	→ 0,235
3	0,365	→ 1,825
4	0,391	→ 1,955
5	0,261	→ 1,305
6	0,131	→ 0,655



$B_6 = 120$ Hz

Mesaj gönderirken Band passfiltreden geçirip yollanabilir.

Bu bant genişliğinde Bir If kullanılabilir.

Örnek Lathi, Pr. 5.2.6

$m(t) = \sin(2000\pi t)$

$k_f = 100\,000$, $k_p = 10$

a) Estimate B_W of $y_{FM}(t)$ and $y_{PM}(t)$

b) Repeat part(a) if the message signal amplitude is doubled

c) " frequency is "

d) Comment on the sensitivity of FM and PM BWs to the spectrum (amplitude) of $m(t)$

$m(t) = \sin(2\pi \underbrace{1000}_{B} t)$, $B = 1000\ Hz$

a) FM: $\Delta f = k_f \cdot mp$ $\rightarrow \Delta f = 100\ khz$

$mp = maks\,|m(t)| = 1$ $\rightarrow \beta = \frac{\Delta f}{B} = 100$

$\rightarrow BG = 2B(\beta+1)$

$= 202\ khz$

PM: $\Delta f = \frac{k_p}{2\pi} \cdot mp'$

$mp' = maks\,|m'(t)| = maks\,|2\pi 100 \underbrace{\cos(\,)}_{1}| = 200\pi$

$\Delta f = \frac{10}{2\pi} \times 200\pi = 10\ khz \rightarrow \beta = \frac{\Delta f}{B} = \frac{10\ khz}{1\ khz} = 10$

$BG = 2B(\beta+1) = 22\ kHz$ (PM)

b) $m(t) = 2\sin(2\pi 1000t)$

FM $mp = 2 \rightarrow \Delta f = k_f mp = 200\ khz \rightarrow \beta = \frac{\Delta f}{B} = \frac{200\ khz}{1\ kHz} = 200$

$\rightarrow BG_{FM} = 2B(\beta+1) = 402\ khz$

PM $mp' = 4000\pi \rightarrow \Delta f = \frac{k_p}{2\pi} \cdot mp' = \frac{10}{2\pi} 4000\pi$

$= 20\ khz$

$\rightarrow \beta = \frac{\Delta f}{B} = 20 \Rightarrow BG_{PM} = 2B(\beta+1) = 42\ kHz$

d) İki modülasyonun da bant genişliği genliğe çok duyarlıdır

c) $m(t) = \sin(2\pi \cdot \underbrace{2000}_{B} t) \Rightarrow B = 2\ kHz$

FM: $mp = 1 \rightarrow \Delta f = k_f mp = 100\ khz$

$\Rightarrow \beta = \frac{\Delta f}{B} = \frac{100k}{2k} = 50 \rightarrow BG_{FM} = 2B(\beta+1) = 204\ khz$

($2k$, 50)

PM: $m_p' = \text{maks}|m(t)| = \text{maks}|2\pi 2000\cos(2\pi 200t)| = 4000\pi$

$\rightarrow \Delta f = \frac{k_p}{2\pi} \cdot m_p' = \frac{10}{2\pi} \cdot 4000\pi = 20\text{kHz}$

$\rightarrow \beta = \frac{\Delta f}{B} = \frac{20k}{2k} = 10 \rightarrow B_{GPM} = 2B(\beta+1)$

$= 2 \times 2k \times (10+1) = 44\text{kHz}$

ol) frekans modülasyonu frekansa çok duyarlı değil ama PM modülasyonu çok duyarlıdır.

Örnek: Lathi Ex 5.5

An angle modulated signal with carrier frequency $f_c = 10^6$ Hz is described by the equation

$y_{EM}(t) = 10\cos[2\pi 10^6 t + 5\sin(3000t) + 10\sin(2000\pi t)]$

$f_1 = \frac{3000}{2\pi}$, $f_2 = \frac{2000}{2} = 1\text{kHz}$

$\phi(t)$, $\theta(t)$

$B_G\{m(t)\} = 1\text{kHz}$

↓ exponential modulation (angle modulation)

a) Find the power of the modulate signal

b) Find the max frequency deviation Δf

c) // the deviation ratio β

d) // the max phase deviation $\Delta\phi$

e) Estimate the BW of $y_{EM}(t)$

$\Delta f = |f_i(t) - f_c|_{\text{maks}}$

a) $P_y = \frac{A_c^2}{2} = 50$ watt

b) $f_i(t) = \frac{1}{2\pi}\frac{d\theta(t)}{dt} = f_c + \frac{1}{2\pi}\frac{d\phi(t)}{dt}$

$\Rightarrow \Delta f = \frac{1}{2\pi}\text{maks}|\dot{\phi}(t)|$

$\Delta f = \frac{1}{2\pi}\text{maks}|5\times 3000\underbrace{\cos(300t)}_{\approx 1} + 10\times 2000\pi\underbrace{\cos(2000\pi t)}_{\approx 1}|$

$\Delta f \cong \frac{1}{2\pi}[15000 + 20000\pi] = 12388\text{ Hz}$

c) $\beta = \frac{\Delta f}{B} = \frac{12350}{1\text{kHz}}$

$$\phi(t) = \begin{cases} k_p m(t) \\ \text{veya} \\ 2\pi k_f \int m(\tau)\,d\tau \end{cases} = 5\sin(3000t) + 10\sin(2000\pi t)$$

$f_1 = \frac{3000}{2\pi} \simeq 477{,}5$ Hz, $f_2 = 1000$

$\Rightarrow BG = f_2 = 1$ kHz

d) $\Delta\phi = \text{maks}\,|\phi(t)| = \text{maks}\,|\underbrace{5\sin(3000t)}_{1} + \underbrace{10\sin(2000\pi t)}_{\approx 1}|$

$\simeq 15$ rad

e) $BG = 2B(\beta+1) = 2 \times 1k \times (12{,}38+1) \simeq 26{,}8$ kHz //

Haftaya 11 de quiz

Periyodik mesaj işaretinin açı modülasyonu

$$y(t) = A_c \cos(2\pi f_c t + \phi(t))$$

$$\phi(t) = \begin{cases} = k_p m(t) & PM \\ = 2\pi k_f \int m(\tau)\,d\tau & FM \end{cases}$$

$m(t)$ $T_m = \frac{1}{f_m}$ periyodik $\Rightarrow$ $\phi(t)$ $T_m = \frac{1}{f_m}$ ile periyodik

$$\hat{y}(t) = A_c e^{j2\pi f_c t} \underbrace{e^{j\phi(t)}}_{x(t)} \rightarrow y(t) = \text{Re}\{\hat{y}(t)\}$$

$$x(t) = e^{j\phi(t)} = \sum_n \underbrace{c_n}_{|c_n| e^{j\angle c_n}} e^{j2\pi n f_m t}, \quad c_n = \frac{1}{T_m}\int_{T_m} \underbrace{e^{j\phi(t)}}_{x(t)} \cdot e^{j2\pi n f_m t}\,dt$$

$$\hat{y}(t) = A_c \cdot e^{j2\pi f_c t} \underbrace{\sum_{n=-\infty}^{+\infty} |c_n| e^{j[(2\pi n f_m t) + \angle c_n]}}_{x(t) = e^{j\phi(t)}}$$

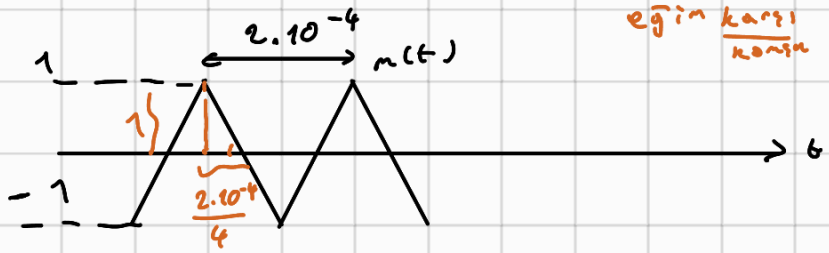
$$y(t) = \text{Re}\{\hat{y}(t)\}$$

$$\boxed{y(t) = A_c \sum_{n=-\infty}^{+\infty} |c_n| \cos(2\pi(f_c + n f_m)t + \angle c_n)}$$

$$\downarrow$$

$$P_y = \frac{A_c^2}{2} \sum_{n=-\infty}^{+\infty} |c_n|^2$$

Örnek Lathi, Ex 5.3



eğim kaçış / komşu $= \frac{1}{\frac{2\times10^{-4}}{4}} = 20000$

a) $BW_{FM} = ?$ $BW_{PM} = ?$

if $k_f = 10^5$, $k_p = 5\pi$

b) Repeat a) if the amplitude of m(t) is doubled

c) Repeat a) if m(t) is time expanded by a factor of 2, i.e, the period of m(t) is 4.10^{-4} s

$T_m = 2.10^{-4}$ s $\rightarrow$ $f_m = \frac{1}{T_m} = 5$ kHz

$$|c_n| = \begin{cases} \frac{4}{\pi^2 n^2} & n \text{ tek} \\ 0 & n \text{ çift} \end{cases}$$

$c_3 = \frac{4}{9\pi^2} = \frac{c_1}{9} = 0,11 c_1$

yani 3. harmonik birincinin %11'i;

$|c_3|^2 = 0,0121 |c_1|^2$ = 1. harmoniğin %1,21'i;

$\rightarrow n = 3$ ise 3. harmonik %11 of the fundamental (f_m) güç katkısı %1,21

$n = 5$ 5. harmonik %4 of the fundamental (f_m) $\rightarrow$ %0,16

Efektif BG'ni $n = 3$. harmonik belirler

$B = 3 f_m = 15$ kHz

a) FM: $\Delta f = k_f \cdot m_p = 100\,000 \times 1 = 100$ kHz $\rightarrow \beta = \frac{\Delta f}{B} = \frac{100k}{15k} = 6,66$

$\Rightarrow BG_{FM} = 2B(\beta + 1) = 2 \times 15k \left(\frac{100}{15} + 1\right)$

$= 230$ kHz

PM: $\Delta f = \frac{k_p}{2\pi} m_p'$, $m_p' = \text{maks}|\dot{m}(t)| = 20000$

$\Rightarrow \Delta f = \frac{5\pi}{2\pi} \times 20000 = 50$ kHz $\rightarrow \beta = \frac{\Delta f}{B} = \frac{50}{15}$

$$BG_{PM} = 2B(\beta+1) = 2\times 15k\left(\frac{50}{15}+1\right)$$
$$= 130\,kHz$$

b) FM $m_p = 2 \rightarrow \Delta f = 200\,kHz \rightarrow BG_{FM} = 2D(\beta+1)$
$$= 2(\underbrace{\Delta f}_{200} + \underbrace{B}_{15})$$
$$= 430\,kHz$$

PM $m_p' = 40000 \rightarrow \Delta f = 100\,kHz \rightarrow BG_{PM} = 2(\Delta f + B)$
$$= 230\,kHz$$

Bant genişliği iki katına çıktı

c) $T_m = 4\cdot 10^{-4} \rightarrow f_m = \frac{1}{T_m} = 2,5\,kHz$

$$BG_{eff} = B = 3f_m = 7,5\,kHz$$

FM: $\Delta f = k_f \cdot m_p = 100\,kHz$

$$\beta = \frac{\Delta f}{B} = \frac{100}{7,5}$$

$$BG_{FM} = 2(\underbrace{\Delta f}_{100k} + \underbrace{B}_{7,5}) = 215\,kHz$$

PM: $\Delta f = \frac{k_p}{2\pi} \cdot m_p'$

$$m_p' = maks\,|m(t)| = 10000$$
$$\Delta f = \frac{5\pi}{2\pi} \cdot 10000 = 25\,kHz$$

$$\rightarrow BG_{PM} = 2(\underbrace{\Delta f}_{25} + \underbrace{B}_{7,5}) = 65\,kHz$$

PM bant genişliği genliğindeki frekans genliğine göre

FM zamandaki genişliğe pek değişmiş oldu!